Shuo Zhang

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Ph.D. Student in Astronomy
Department of Astronomy, Shanghai Jiao Tong University
Shanghai, China

Research Interests

Strong gravitational lensing; Lensed quasars; Quasar microlensing; Galaxy mass distributions; Stellar and dark matter components; Hierarchical Bayesian inference.

Education

- **Shanghai Jiao Tong University** *2024 – Present*
  Ph.D. Student in Astronomy, Department of Astronomy, School of Physics and Astronomy Shanghai, China
  - Advisor: Prof. Alessandro Sonnenfeld
  - Research focus: statistical strong lensing, lensed quasars, microlensing, and galaxy mass distributions
- **Xiamen University** *2020 – 2024*
  B.S. in Astronomy Xiamen, China
  - Advisor: Prof. Taotao Fang
  - Undergraduate research on FAST H I data analysis and Galactic high-velocity cloud mapping

Research Experience

- **Statistical Strong-Lensing Analysis of Gaia Multi-Peak Lenses** *2026 – Present*
  Ph.D. Research, Shanghai Jiao Tong University Shanghai, China
  - Analyze lensed quasars selected with the Gaia Multi-Peak (GMP) method.
  - Develop statistical methods to infer lens-galaxy mass distributions from lensing observables.
  - Investigate selection effects in GMP-selected lens samples and their impact on population-level inference.
- **Statistical Strong Lensing with Microlensed Quasars** *2024 – 2026*
  Ph.D. Research, Shanghai Jiao Tong University Shanghai, China
  - Developed hierarchical Bayesian methods for population-level inference with lensed quasars.
  - Focused on separating stellar mass components from dark matter density profiles in strong-lens galaxies.
  - Analyzed statistical microlensing effects in lensed quasars.
- **FAST H I Data Analysis and High-Velocity Clouds** *2023 – 2024*
  Undergraduate Research, Xiamen University Xiamen, China
  - Processed and analyzed large-area FAST H I survey data.
  - Produced maps and measurements of Galactic high-velocity cloud structures.
  - Compared FAST measurements with archival H I survey data.
  - Developed Python-based workflows for spectral-line data processing and visualization.

Publications

[1]] Zhang, S., et al. **Statistical Strong Lensing. V. Inference with Microlensed Quasars**. Manuscript in preparation.

Presentations

- **Poster Presentation** *Year*
  Guo Shoujing Astronomy Meeting
  - Presented research related to strong gravitational lensing and lensed quasars.

Honors and Awards

- **Tsung-Dao Lee Ph.D. Program** *2024 – 2026*
  Tsung-Dao Lee Institute, Shanghai Jiao Tong University
- **Outstanding Graduate** *2024*
  Xiamen University
- **Shanghai Astronomical Observatory Scholarship** *2021, 2022, 2023*
  Shanghai Astronomical Observatory

Technical Skills

- **Programming:** Python, Shell scripting, Git, LaTeX
- **Scientific Computing:** NumPy, SciPy, Astropy, Matplotlib, Jupyter
- **Astronomical Data Analysis:** HST imaging, JWST data, MUSE/IFU data, FAST H I data
- **Strong Lensing and Inference:** Strong-lens modeling, microlensing statistics, hierarchical Bayesian inference, MCMC
- **Computing:** Linux, HPC environments, remote servers, Gravity cluster administration

Additional Information

Languages: Chinese (native), English (working proficiency)
Research Tools: Python-based astronomical analysis, Bayesian statistical modeling, LaTeX scientific writing

References

Available upon request.